# Aranyabodh Annual Operating Cost Projection

## November 2026 to October 2027

Revised 17 October 2026 | All amounts in INR | USD 1 = INR 100

## Executive summary

The recommended annual operating budget is INR 9.00 lakh for 1,000 registered users. Expected recurring expenditure is INR 7.54 lakh, including a provisional tax allowance. A 15% contingency and INR 10,000 for initial setup bring the calculated funding requirement to INR 8.77 lakh. The recommended budget leaves INR 23,059 in additional headroom.

| Annual funding summary | INR |
|---|---|
| Recurring services and operating allowances before tax | 6,38,866 |
| Provisional tax allowance at 18% | 1,14,996 |
| Recurring expenditure including tax allowance | 7,53,862 |
| Contingency at 15% | 1,13,079 |
| Initial setup allowance | 10,000 |
| Funding required including contingency and setup | 8,76,941 |
| **Recommended annual allocation** | **9,00,000** |

### Usage and planning assumptions

| Planning input | Base assumption |
|---|---|
| Registered users | 1,000 forest department officers and commercial users |
| Monthly active users | 600 users |
| Average daily active users | 200 users |
| Questions per active user daily | 4.5 questions |
| Question volume | 27,000 in a 30-day month and 3,28,500 over the year |
| Initial document library | 500 documents averaging 20 pages each |
| Monthly document additions | 100 documents averaging 20 pages each |
| Exchange rate | INR 100 per USD across all USD-priced items |

The forecast assumes 600 monthly active users, each using the app on about 10 days in a typical month. This gives an average of 200 daily active users. These assumptions suit an intermittently used reference tool and should be checked against actual usage. The user base remains constant, and monthly volumes follow the calendar.

The budget covers infrastructure, AI APIs, storage, backups, messaging, monitoring and business subscriptions. It excludes development, maintenance labour, salaries, manual document review and sales labour. Subscription revenue is not deducted from operating costs.

# 1 Operating cost breakdown

The monthly comparison uses 30 days; the annual total uses actual calendar days. Amounts are before the provisional tax allowance. USD-priced services use INR 100 per USD, while fixed rupee allowances remain unchanged.

| Cost category | Nov Dec 2026 | Jan Oct 2027 | Annual total |
|---|---|---|---|
| AI generation | 23,100 | 42,158 | 4,74,166 |
| Main app service including document processing | 2,500 | 2,500 | 30,000 |
| Supabase Pro and Small compute | 3,000 | 3,000 | 36,000 |
| Redis Cloud at USD 6 monthly | 600 | 600 | 7,200 |
| Storage backups and bandwidth | 1,000 | 1,000 | 12,000 |
| Monitoring logging and error tracking | 1,000 | 1,000 | 12,000 |
| OCR and embeddings | 500 | 500 | 6,000 |
| Email SMS and WhatsApp | 2,000 | 2,000 | 24,000 |
| Domain renewal allocation | 125 | 125 | 1,500 |
| Business and support subscriptions | 2,000 | 2,000 | 24,000 |
| Optional Jev provision | 1,000 | 1,000 | 12,000 |
| Static frontend DNS CDN and TLS | 0 | 0 | 0 |
| **Operating subtotal** | **36,825** | **55,883** | **6,38,866** |

### Hosting and document processing

The main app service is budgeted at USD 25 per month. The API and document-processing worker share this allocation, whether they run in one container or separate containers on the same VM. The budget assumes enough CPU and memory for both. An additional paid instance would increase hosting costs.

Supabase costs USD 30 per month: USD 25 for Pro, plus USD 15 for Small compute, less the USD 10 compute credit. Redis Cloud is budgeted at USD 6 per month. Database usage is assumed to remain within the included disk and transfer limits.

### Operating allowances

A single INR 1,000 monthly allowance covers monitoring, logging, error tracking and uptime checks. Included platform features should be used first; this allowance covers any gaps or overages.

Storage allows for about 100 GB of originals and backup copies. Messaging assumes 1,000 SMS, 4,000 WhatsApp utility messages and 6,000 emails per month, including provider fees. A separate INR 2,000 allowance covers business email, billing and support tools. These amounts are spending limits rather than committed subscriptions.

Jev is optional and disabled by default. The INR 1,000 monthly provision allows for approximately INR 907 at 27,000 questions, assuming 8,000 decision-input tokens per question at USD 0.042 per million tokens. Gemini costs are not reduced to offset it. The provision remains unspent if Jev is not enabled.

## 2 Basis of the AI estimate

Each question may require several model calls for rewriting, planning, answering and verification. The estimate includes these stages and billable reasoning tokens. It incorporates the announced January increase in Gemini Flash rates and holds other provider rates constant.

| Model USD per million tokens | Input | Output |
|---|---|---|
| Gemini 3.5 Flash Lite throughout the year | 0.30 | 2.50 |
| Gemini 3.8 Flash through December 2026 | 0.75 | 3.75 |
| Gemini 3.8 Flash from January 2027 | 1.50 | 7.50 |

| Question class | Share | 2026 INR | 2027 INR |
|---|---|---|---|
| Simple | 25% | 0.1111 | 0.1111 |
| Procedural | 50% | 0.6667 | 1.2000 |
| Complex | 25% | 1.6667 | 3.1667 |

The table estimates generation costs before recovery attempts, using INR 100 per USD. Simple questions use Flash Lite. Flash is assumed to account for 80% of procedural costs and 90% of complex-question costs; those components double in January. The class mix and costs are planning estimates informed by selected project evaluations and should be validated against live traffic.

### Calculation method

The weighted question cost includes a single 10% allowance for recovery attempts. This gives INR 0.85556 per question through December 2026 and INR 1.56139 from January 2027, before tax.

A typical month produces 27,000 questions: 200 daily active users × 4.5 questions × 30 days. Estimated generation cost is INR 23,100 through December and INR 42,158 from January. No caching, batch or promotional discounts are assumed.

### OCR and embeddings

Half of the 2,000 new pages added each month are assumed to require optical character recognition (OCR). Text indexing allows 600 tokens per page and 20% chunk overlap, or 1.44 million document tokens. Query embeddings allow 200 input tokens per question.

At INR 100 per USD, a typical month costs about INR 150 for OCR, INR 29 for document embeddings and INR 108 for query embeddings. The combined INR 287 fits within the INR 500 allowance. Rates are USD 1.50 per 1,000 OCR pages and USD 0.20 per million text-embedding tokens; OCR free allowances are not deducted.

### Tax and contingency

An 18% tax allowance is applied to the operating subtotal without deducting possible input credits. Actual tax treatment will follow supplier invoices. Contingency is 15% of recurring expenditure including this allowance. It covers currency movement, overages and estimation uncertainty and is held as a reserve.

# 3 Monthly expenditure and funding

The projection runs from 1 November 2026 to 31 October 2027. Other recurring costs are INR 13,725 per month. Expenditure includes the provisional tax allowance and excludes contingency and initial setup. All monetary columns are in INR.

| Month | Questions | AI cost | Other costs | Tax | Expenditure |
|---|---|---|---|---|---|
| Nov 2026 | 27,000 | 23,100 | 13,725 | 6,629 | 43,454 |
| Dec 2026 | 27,900 | 23,870 | 13,725 | 6,767 | 44,362 |
| Jan 2027 | 27,900 | 43,563 | 13,725 | 10,312 | 67,600 |
| Feb 2027 | 25,200 | 39,347 | 13,725 | 9,553 | 62,625 |
| Mar 2027 | 27,900 | 43,563 | 13,725 | 10,312 | 67,600 |
| Apr 2027 | 27,000 | 42,158 | 13,725 | 10,059 | 65,941 |
| May 2027 | 27,900 | 43,563 | 13,725 | 10,312 | 67,600 |
| Jun 2027 | 27,000 | 42,158 | 13,725 | 10,059 | 65,941 |
| Jul 2027 | 27,900 | 43,563 | 13,725 | 10,312 | 67,600 |
| Aug 2027 | 27,900 | 43,563 | 13,725 | 10,312 | 67,600 |
| Sep 2027 | 27,000 | 42,158 | 13,725 | 10,059 | 65,941 |
| Oct 2027 | 27,900 | 43,563 | 13,725 | 10,312 | 67,600 |
| **Annual total** | **3,28,500** | **4,74,166** | **1,64,700** | **1,14,996** | **7,53,862** |

### Funding allocation

| Period | Monthly allocation | Period total |
|---|---|---|
| November 2026 including initial allowance | 65,000 | 65,000 |
| December 2026 | 55,000 | 55,000 |
| January to October 2027 each month | 78,000 | 7,80,000 |
| **Annual budget** | | **9,00,000** |

The INR 9,00,000 allocation covers INR 7,53,862 of recurring expenditure, INR 1,13,079 of contingency and INR 10,000 for initial setup. A further INR 23,059 remains as headroom. Average recurring expenditure is INR 62,822 per month.

Calendar length explains variation within each pricing period. The January step reflects the Flash rate increase. Figures are calculated at full precision and rounded to the nearest rupee, so displayed rows may differ slightly from totals.

# 4 Scenarios and scope

The scenarios vary daily active users while keeping the registered population at 1,000, use at 4.5 questions per active day and other allowances unchanged. Higher activity may also require more hosting capacity.

| Scenario | DAU | Annual expenditure | Funding including reserve and setup |
|---|---|---|---|
| Low | 100 | 4,74,104 | 5,55,220 |
| Base | 200 | 7,53,862 | 8,76,941 |
| High | 350 | 11,73,499 | 13,59,524 |

### Initial provisioning allowance

| Initial setup item | INR |
|---|---|
| Initial document OCR and embeddings including tax allowance | 1,100 |
| Conditional sender registration or messaging onboarding | 6,000 |
| Other initial vendor fees and provisioning headroom | 2,900 |
| **Total initial allowance** | **10,000** |

Initial OCR and embeddings for the assumed 10,000-page library cost about INR 894 before tax, or INR 1,055 including the tax allowance. Messaging onboarding and other setup amounts are conditional provisions, not committed fees. Domain renewal is already included in recurring costs.

### Costs outside this operating scope

This report covers app operating costs. Development and maintenance labour, salaries, manual document review, and sales and marketing labour remain outside the budget.

Payment processing depends on collections. If subscriptions are sold online, allow approximately 3% of gross receipts separately and replace this estimate with the selected payment arrangement. No revenue is assumed or deducted here.

Paid content licences, formal security audits, enterprise support, restricted hosting and contractual availability guarantees require separate budgets if needed. Advanced extraction, web search, voice, imagery and managed point-in-time database recovery are also outside this estimate.

### Budget review

Review usage and invoices monthly. Reassess hosting if document processing slows answers, monthly questions exceed 30,000, or generation cost per question exceeds the estimate by 20%. Replace provisional allowances with actual bills as they become available.